



MYTHIC

Compiler Optimization Report - BEVFormer-Tiny

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1 Summary

For BEVFormer-Tiny at 6x928x1600x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 31.93 ms, of which
- 27.30 ms is Analog NPU processing time, which corresponds to
 - 68.5% ACE utilization, using 72 Analog Compute Engines; and of which
- 4.63 ms is Digital NPU processing time,
 - using 288 Digital NPU cores, which is sufficient to achieve 30 fps.

The number of used Digital NPU cores (288) was selected to achieve at least 30 fps. If more Digital NPU cores were used, the Digital NPU processing time could be shortened, but the benefit would be small because the Analog NPU processing time already accounts for most of the total processing time.

There is a slight digital bottleneck, due to SRAM Read/Write time, which increases the Analog NPU processing time by ~4%. This will be addressed in the future by improving compiler modeling/optimization of SRAM accesses. A reduction in SRAM Read/Write time by ~30% should be possible, but only ~4% is needed.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 68.5% is already fairly high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 4.4 Watts at the target frame rate (30 fps), and
- 4.5 Watts at the maximum frame rate (31 fps), if restricted to using 288 Digital NPU cores.

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the “high_optimization” compiler config for BEVFormer-Tiny.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

Because the target frame rate (30 fps) is so close to the maximum frame rate (31 fps), we report details only for the target frame rate (30 fps). Reproducing the results for 31 fps would be similar.

The `mythic-compiler` command was:

```
mythic-compiler \  
  --input-artifact $MYTHIC_SDK_ROOT/models/training/bevformer/bevformer-tiny-1600x900-  
trained.tar.gz \  
  --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/bevformer/compiler/  
bevformer_tiny_6x928x1600x3_m2072_high_optimization.yaml \  

```

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```
--output-artifact /rscratch/tonbomythic3/<your_username>/
bevformer_tiny_6x928x1600x3_m2072_high_optimization_compiled.tar.gz
```

The first `mythic-ppa-estimators` command for the target frame rate (30 fps) was:

```
mythic-ppa-estimators \
  --estimate-performance \
  --estimate-power \
  /rscratch/tonbomythic3/<your_username>/
bevformer_tiny_6x928x1600x3_m2072_high_optimization_compiled.tar.gz
```

To speed up simulation time and to reduce simulation memory usage, the above commands estimate performance and power assuming 1-camera input (1x928x1600x3) for the Analog NPU part of the estimation, and assuming 6-camera input (6x928x1600x3) for the Digital NPU part of the estimation.

So, to correctly determine the power for the Analog NPU part of the network for 6-camera input, the `mythic-ppa-estimators` command must be rerun using 6 times the inference rate ($6 \times 30 = 180$ fps). In other words, running 1-camera input at 180 fps is the same as running 6-camera input at 30 fps, because the Analog NPU processes the camera inputs sequentially.

So, the second `mythic-ppa-estimators` command, using 180 fps to correctly estimate Analog NPU power, was:

```
mythic-ppa-estimators \
  --estimate-performance \
  --estimate-power \
  --power-inference-rate 180 \
  /rscratch/tonbomythic3/<your_username>/
bevformer_tiny_6x928x1600x3_m2072_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 4.55 ms
Analog NPU Estimated Frame Rate for Compiled ONNX File: 219.55 fps (frames per second)
Critical Path ACE Latency: 4.37 ms
Maximum SRAM Read/Write Time (over 18 ACE tiles): 4.55 ms
Maximum SIMD Operation Time (over 18 ACE tiles): 0.00 ms
Average SRAM Read/Write Time (averaged over 18 ACE tiles): 3.49 ms
Average SIMD Operation Time (averaged over 18 ACE tiles): 0.00 ms
Total Executed ACE Operations: 1,403,600
Total Executed ACE MACs: 159,170,560,000
```

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Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 224.58 ms
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 72 ACEs): 3.12 ms

ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 68.48%

Total SRAM Bytes Read Across Chip: 4,328,955,464 bytes

Total SRAM Bytes Written Across Chip: 2,938,200,936 bytes

Estimated Die Area to Achieve Estimated Processing Time: 379.64 mm²

Digital NPU Performance Metrics (from separate ONNX graph processing)

Total Estimated MACs Targeting Digital NPU: 16,529,000,000

Digital NPU MAC Utilization: 9.69%

Digital NPU Cores: 288

Digital NPU Frequency: 2,000 MHz

Digital Estimated Frame Processing Time: 4.63 ms

Digital Estimated Frame Rate: 216.15 fps (frames per second)

Combined Analog + Digital NPU Latency: 9.18 ms

Combined Analog + Digital NPU Total Estimated Frame Rate: 108.92 fps (frames per second)

NOTE: The information reported above may include the Analog NPU or the Analog NPU and a digital NPU as indicated. This estimated processing time and associated frame rate only covers the execution of the compiled ONNX model(s) running on one or both of those compute solutions in isolation. It does not include PCIe transfer overhead. Performance estimation for additional processing steps and a combined compute hardware solution will be added in future versions.

Because the above reports the Analog NPU processing time for the 1-camera version of BEVFormer, the total Analog NPU processing time for 6 cameras is $6 * 4.55 \text{ ms} = 27.30 \text{ ms}$.

Together with the digital latency, the total latency is $27.30 \text{ ms} + 4.63 \text{ ms} = 31.93 \text{ ms}$.

The maximum frames per second is $1 \text{ frame} / 31.93 \text{ ms} = 31.32 \text{ fps}$.

4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimates. Comments have been added to the output snippets below to show which power values to use and which to ignore.

Power for the Digital NPU part of the network, at the target frame rate (30 fps), is determined from the following `mythic-ppa-estimators` output:

```
Process Nodes: Analog 28nm, Digital 5nm
Number of ACEs: 24
Number of Digital NPU Cores: 288
Inference Rate: 30 frames/second (fps)    <--- Use 30 fps to estimate Digital NPU power
```

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```
Analog NPU Estimated Power: 0.580 W for target ONNX file    <--- Ignore
  Functional Unit Power: 0.436 W
  Interconnect Power:    0.144 W

Digital NPU Estimated Power: 1.218 W for digital target ONNX file    <--- Use this for
Digital NPU power

Total Combined (Analog + Digital NPU) Power: 1.797 W    <--- Ignore

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

The Analog NPU part of the network must run at effectively six times the target frame rate ($6 * 30 \text{ fps} = 180 \text{ fps}$), so its power is determined from the following `mythic-ppa-estimators` output:

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs:  24
Number of Digital NPU Cores: 288
Inference Rate: 180 frames/second (fps)    <--- Use 180 fps to estimate Analog NPU power

Analog NPU Estimated Power: 3.160 W for target ONNX file    <--- Use this for Analog NPU
power
  Functional Unit Power: 2.296 W
  Interconnect Power:    0.864 W

Digital NPU Estimated Power: 7.307 W for digital target ONNX file    <--- Ignore

Total Combined (Analog + Digital NPU) Power: 10.467 W    <---- Ignore

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

The power required is 1.218 W for the Digital NPU part and 3.160 W for the Analog NPU part, for a combined total power of 4.378 W at the target frame rate (30 fps).

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
===== L0 IR stats =====  
      Tiles: 19  
      Buffers: 322  
      Launchers: 520  
    ACE Launchers: 235  
      ACE Weights: 35756800  
    ACE calculations: 1403600  
    Max ACE calculations: 82476 on tile 2_5  
=====
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

Summary:

- 81.93% ACE utilization is predicted by the compiler,
- assuming no digital bottlenecks (which can typically be eliminated).
- 1,403,600 ACE calculations are spread across 72 ACEs, which is
- 19,494.44 ACE calculations per ACE (on average).
- 23,793 ACE calculations are in the critical path, implying
- 3.81 ms predicted inference time (160 ns per ACE calculation).

Tile	ACEs	ACE calculations	ACE calculations Per ACE	Predicted Full Network ACE Critical Path	Predicted ACE Utilization
1_1	0	N/A	N/A	N/A	N/A
3_1	4	73,544	18,386.00	23,793	77.27%
4_1	4	80,910	20,227.50	23,793	85.01%
1_2	4	74,994	18,748.50	23,793	78.80%
2_2	4	75,574	18,893.50	23,793	79.41%
3_2	4	76,212	19,053.00	23,793	80.08%
4_2	4	79,054	19,763.50	23,793	83.06%
1_3	4	81,432	20,358.00	23,793	85.56%
2_3	4	69,890	17,472.50	23,793	73.44%
3_3	4	81,896	20,474.00	23,793	86.05%
4_3	4	79,460	19,865.00	23,793	83.49%
1_4	4	80,040	20,010.00	23,793	84.10%
2_4	4	77,604	19,401.00	23,793	81.54%

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3_4	4	81,838	20,459.50	23,793	85.99%
4_4	4	76,096	19,024.00	23,793	79.96%
1_5	4	77,662	19,415.50	23,793	81.60%
2_5	4	82,476	20,619.00	23,793	86.66%
3_5	4	73,254	18,313.50	23,793	76.97%
4_5	4	81,664	20,416.00	23,793	85.81%
All	72	1,403,600	19,494.44	23,793	81.93%

5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

Summary:

- 39.88% of analog weight storage capacity is used, using
- 35,756,800 weights out of 89,653,248 capacity, using
- 235 weight blocks.

Tile	Weight Blocks	Allocated Weights	Weight Capacity	Utilization
3_1	14	2,392,704	4,980,736	48.04%
4_1	15	2,005,248	4,980,736	40.26%
1_2	13	2,130,688	4,980,736	42.78%
2_2	12	1,551,488	4,980,736	31.15%
3_2	13	1,633,152	4,980,736	32.79%
4_2	12	1,502,208	4,980,736	30.16%
1_3	12	1,911,936	4,980,736	38.39%
2_3	12	1,731,712	4,980,736	34.77%
3_3	12	1,647,744	4,980,736	33.08%
4_3	12	1,698,944	4,980,736	34.11%
1_4	13	1,622,784	4,980,736	32.58%
2_4	14	2,736,768	4,980,736	54.95%
3_4	14	2,226,944	4,980,736	44.71%
4_4	12	1,535,104	4,980,736	30.82%
1_5	14	2,589,312	4,980,736	51.99%
2_5	13	2,294,528	4,980,736	46.07%
3_5	13	2,130,688	4,980,736	42.78%

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4_5	15	2,414,848	4,980,736	48.48%
All	235	35,756,800	89,653,248	39.88%

5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

Summary:

- 5,923,431 bytes of SRAM is used, consisting of
- 5,178,528 bytes **for** buffers (87.42%) and
- 744,903 bytes **for** control flow (12.58%), using
- 7.85% of the simulated on-chip SRAM capacity.

Tile	Buffer Allocation (bytes)	Control Flow Allocation (bytes)	Total Allocation (bytes)	SRAM Capacity (bytes)	Utilization
3_1	51,712	39,357	91,069	4,194,304	2.17%
4_1	4,288	17,820	22,108	4,194,304	0.53%
1_2	265,408	34,507	299,915	4,194,304	7.15%
2_2	438,592	50,069	488,661	4,194,304	11.65%
3_2	310,240	36,744	346,984	4,194,304	8.27%
4_2	351,808	23,217	375,025	4,194,304	8.94%
1_3	532,288	25,827	558,115	4,194,304	13.31%
2_3	407,424	58,002	465,426	4,194,304	11.10%
3_3	349,568	52,264	401,832	4,194,304	9.58%
4_3	330,240	37,547	367,787	4,194,304	8.77%
1_4	244,992	40,147	285,139	4,194,304	6.80%
2_4	247,424	38,300	285,724	4,194,304	6.81%
3_4	616,448	98,918	715,366	4,194,304	17.06%
4_4	537,600	52,815	590,415	4,194,304	14.08%
1_5	111,616	20,834	132,450	4,194,304	3.16%

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2_5	210,240		51,535		261,775		4,194,304		6.24%	
3_5	164,864		42,818		207,682		4,194,304		4.95%	
4_5	3,776		24,182		27,958		4,194,304		0.67%	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+										
All	5,178,528		744,903		5,923,431		75,497,472		7.85%	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+										